VAJA-07 — Varno deljenje spletne aplikacije s podatkovno bazo v AWS

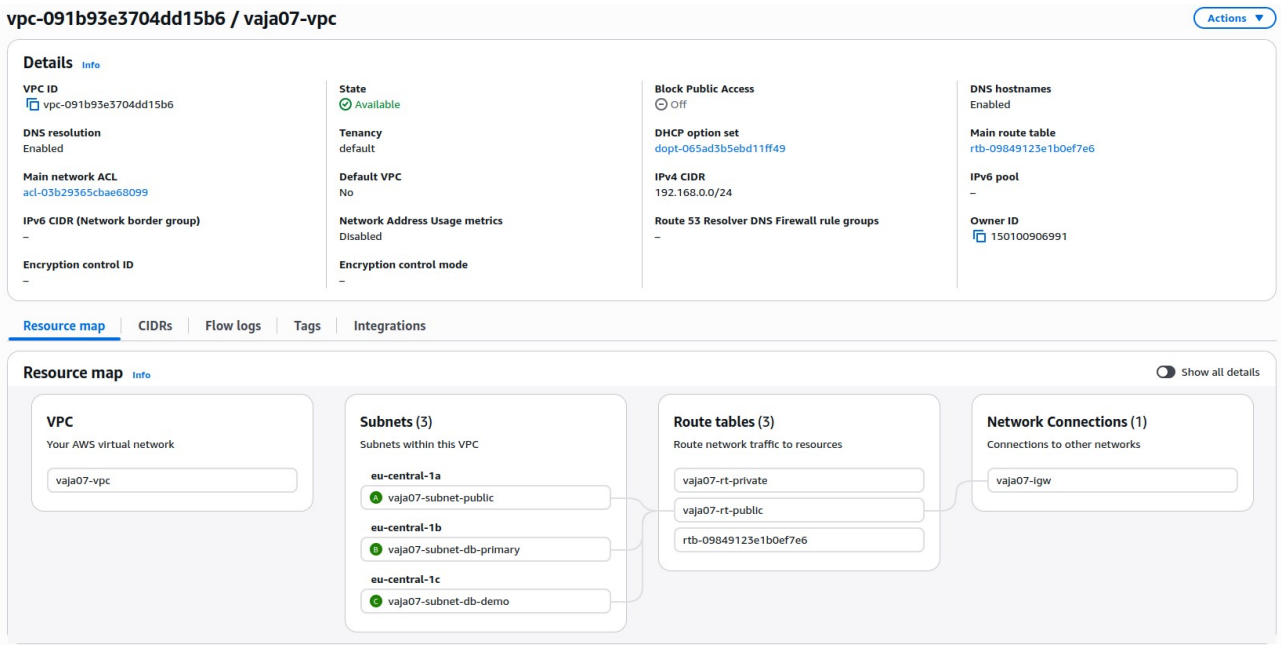
PREDPOGOJI

- AWS račun z aktivnim IAM uporabnikom (admin pravice)
- Regija `eu-central-1`
- Lokalno nameščeni: `aws` CLI, `ssh`, `scp`, `curl`, brskalnik
- Lokalno nameščen klient `mariadb` (za morebitne oddaljene teste)

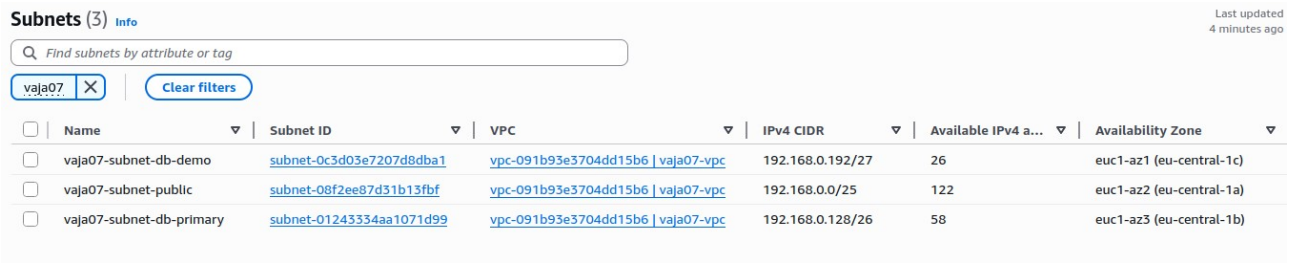
POSTOPEK

1. Omrežje

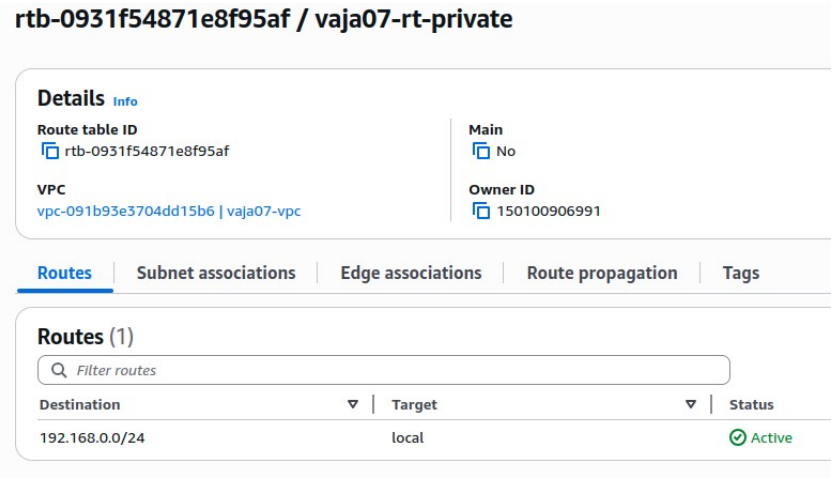
Z zagonom `bash create-vaja07-infra.sh` sem ustvaril nov VPC vaja07-vpc (vpc-091b93e3704dd15b6) z IPv4 CIDR 192.168.0.0/24, kar pokriva 256 naslovov oziroma 254 uporabnih hostov. V VPC sem dodal tri podomrežja, vsako v svoji Availability Zone, s takšnimi maskami, da ustrezajo zahtevanemu številu računalnikov.

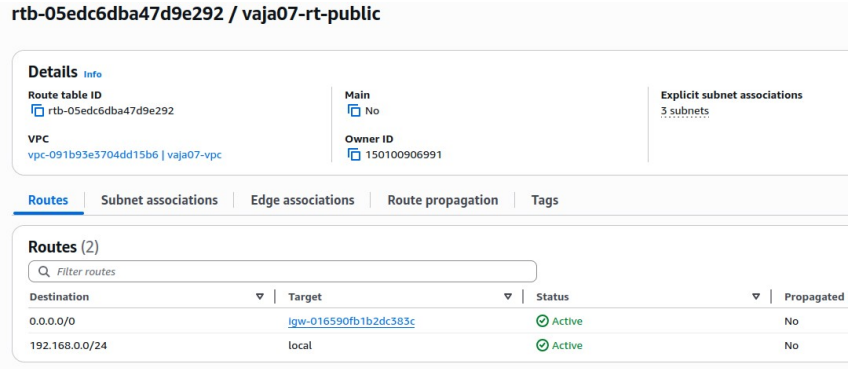


Podomrežje vaja07-subnet-public (subnet-08f2ee87d31b13fbf) pokriva 192.168.0.0/25 (128 naslovov, za 126 hostov) in leži v eu-central-1a; namenjeno je spletnemu strežniku. Privatno podomrežje vaja07-subnet-db-primary (subnet-01243334aa1071d99) pokriva 192.168.0.128/26 (64 naslovov) v eu-central-1b, medtem ko je vaja07-subnet-db-demo (subnet-0c3d03e7207d8dba1) v eu-central-1c z CIDR 192.168.0.192/27 (32 naslovov). Ločene AZ zagotavljajo odpornost na izpad posamezne podatkovne lokacije.



Ustvaril sem Internet Gateway vaja07-igw (igw-016590fb1b2dc383c) in ga priklopil na VPC. Nato sem pripravil dve usmerjevalni tabeli: vaja07-rt-public (rtb-05edc6dba47d9e292) s privzeto potjo 0.0.0.0/0 proti IGW in vaja07-rt-private (rtb-0931f54871e8f95af) brez privzete poti, ki ima le lokalno VPC pot. Med postavitvijo so vsa tri podomrežja začasno asociirana s public RT, saj morajo EC2-2 in EC2-3 pri namestitvi programske opreme prek `apt` doseči internet; po končani namestitvi se Sub2 in Sub3 asociirata na privatno RT, s čimer postaneta resnično privatni.





2. Varnostne skupine in EC2 instance

Za vsako plast aplikacije sem ustvaril ločeno varnostno skupino. Spletni nivo uporablja vaja07-sg-web (sg-0b0c8dc0cc56a0f77), ki dovoljuje TCP 22 in TCP 80 iz 0.0.0.0/0. Podatkovni nivo uporablja vaja07-sg-db (sg-06c0a2eccbb5a6f17), ki dovoljuje TCP 22 in TCP 3306 samo iz vira sg-vaja07-sg-web — torej samo spletni strežnik dostopa do baze, direkten dostop iz interneta ni mogoč.

sg-0b0c8dc0cc56a0f77 - vaja07-sg-web

Details

Security group name	Security group ID	Description	VPC ID
vaja07-sg-web	sg-0b0c8dc0cc56a0f77	VAJA-07 web tier: SSH + HTTP	vpc-091b93e3704dd15b6

Owner	Inbound rules count	Outbound rules count
150100906991	2 Permission entries	1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (2)

Manage tags

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
–	sgr-0f45566ae95742d39	IPv4	SSH	TCP	22	0.0.0.0/0	–
–	sgr-0e99d5f5f6b95cd44	IPv4	HTTP	TCP	80	0.0.0.0/0	–

sg-06c0a2eccbb5a6f17 - vaja07-sg-db

Details

Security group name	Security group ID	Description	VPC ID
vaja07-sg-db	sg-06c0a2eccbb5a6f17	VAJA-07 db tier: MariaDB + bastion SSH iz sg-web	vpc-091b93e3704dd15b6

Owner	Inbound rules count	Outbound rules count
150100906991	2 Permission entries	1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (2)

Manage tags

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
–	sgr-08a484b8251f816ac	–	SSH	TCP	22	sg-0b0c8dc0cc56a0f77 / vaja07-sg-web	–
–	sgr-00fb7ec3bdfd882af	–	MYSQL/Aurora	TCP	3306	sg-0b0c8dc0cc56a0f77 / vaja07-sg-web	–

Nato sem ustvaril skupni ključni par vaja07-key in shranil zasebni ključ v `vaja07-key.pem`. Z njim sem zagnal tri EC2 instance Debian 12 tipa `t3.micro`. Instanca vaja07-ec2-web (i-0f1345d69f0c9355c) teče v Sub1 in ima pripeto Elastic IP 63.184.14.98 ter privatni naslov 192.168.0.8. Instanca vaja07-ec2-db-primary (i-0a48e5b0363281bf1) teče v Sub2 s privatnim naslovom 192.168.0.163, instanca vaja07-ec2-db-demo (i-0362a258d57feeccc) pa v Sub3 s privatnim naslovom 192.168.0.198.

Instances (3) Info

Connec

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

vaja07

Clear filters

Name	Instance ID	Instance state	Public IPv4 ad...	Private IP address	Availability Zone
vaja07-ec2-db-demo	i-0362a258d57feeccc	Running	63.179.58.89	192.168.0.198	eu-central-1c
vaja07-ec2-db-primary	i-0a48e5b0363281bf1	Running	3.122.192.26	192.168.0.163	eu-central-1b
vaja07-ec2-web	i-0f1345d69f0c9355c	Running	63.184.14.98	192.168.0.8	eu-central-1a

Pri zagonu EC2-3 sem uporabil `--user-data` parameter, ki pošlje skripto `user-data-ec2-db-demo.sh` na instanco; cloud-init jo zažene ob prvem zagonu in samodejno namesti MariaDB. S tem sem demonstriral namestitev strežnika že pri izdelavi EC2.

3. Spletni strežnik (EC2-1)

Z zasebnim ključem `vaja07-key.pem` sem se prek SSH povezal na admin@63.184.14.98 in preveril, da je operacijski sistem Debian 12 pripravljen. Sistem je odgovoril z imenom ip-192-168-0-8 ter zasebnim naslovom 192.168.0.8, kar potrjuje, da instanca teče v Sub1.

```
urban-ambrozic@urban-ambrozic-ubuntu:~$ ssh -i /tmp/vaja07-key.pem admin@63.184.14.98
Linux ip-192-168-0-8 6.1.0-44-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.164-1 (2026-03-09) x86_64

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the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
admin@ip-192-168-0-8:~$ whoami && uname -a && hostname -I
admin
Linux ip-192-168-0-8 6.1.0-44-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.164-1 (2026-03-09) x86_64 GNU/Linux
192.168.0.8
```

Nato sem z ukazom `scp` na EC2-1 prenesel `setup-web-server.sh` in datoteke aplikacije (`index.html`, `vstavi.php`, `izpis.php`, `db.php`). V `db.php` sem pred prenosom zamenjal spremenljivko `$host` z zasebnim naslovom EC2-2 (192.168.0.163), saj bo spletni strežnik do baze dostopal preko zasebnega VPC omrežja.

Na EC2-1 sem zagnal `sudo bash setup-web-server.sh`, ki je z `apt` namestil `apache2`, `php`, `php-mysqli` in `mariadb-client`, omogočil Apache servis (`systemctl enable --now apache2`) ter kopiral datoteke aplikacije v `/var/www/html/`. Po končani namestitvi sem iz lokalnega terminala z `curl http://63.184.14.98/` preveril delovanje — odziv je bil HTTP 200 s HTML obrazcem za vnos elementa in količine, kar potrjuje, da Apache teče in pravilno streže aplikacijo.

4. Glavni podatkovni strežnik (EC2-2)

Ker je EC2-2 v zasebnem podomrežju in do njega ne moremo dostopati neposredno iz interneta, sem se nanjo povezal prek EC2-1 kot bastion. Z `scp` sem na EC2-2 prenesel `setup-db-server.sh` in `setup-db.sql`, nato pa prek SSH zagnal `sudo bash setup-db-server.sh`. Skripta je z `apt` namestila `mariadb-server`, v `/etc/mysql/mariadb.conf.d/50-server.cnf` nastavila `bind-address = 0.0.0.0`, z `systemctl enable --now mariadb` omogočila servis ter uvozila `setup-db.sql`.

SQL skripta je ustvarila bazo AlmaMater, v njej tabelo nakup s stolpci id, element in kolicina, uporabnika urban z admin pravicami nad to bazo ter vstavila testna vnosa (kruh, 2) in (mleko, 1). Po končani namestitvi sem se prek bastiona prijavil na EC2-2 in z `systemctl status mariadb --no-pager` preveril, da je servis aktiven in teče.

```
admin@ip-192-168-0-8:~$ ssh -i ~/vaja07-key.pem admin@192.168.0.163
The authenticity of host '192.168.0.163 (192.168.0.163)' can't be established.
ED25519 key fingerprint is SHA256:NO4C/CGAxyAN/1i4aTUT7T8gq+kwJisq7WjeEtKkhf0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.163' (ED25519) to the list of known hosts.
Linux ip-192-168-0-163 6.1.0-44-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.164-1 (2026-03-09) x86_64

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individual files in /usr/share/doc/*/copyright.

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permitted by applicable law.
admin@ip-192-168-0-163:~$ systemctl status mariadb --no-pager
● mariadb.service - MariaDB 10.11.14 database server
     Loaded: loaded (/lib/systemd/system/mariadb.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-04-20 18:32:30 UTC; 3min 15s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 1816 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /var/run/mysqld (code=exited, status=0/SUCCES
S)
    Process: 1817 ExecStartPre=/bin/sh -c systemctl unset-environment _WSREP_START_POSITION (code=exited, status=0/SUCCE
SS)
    Process: 1819 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera_recovery ] && VAR= ||   VAR=`/usr/bin/galera_recovery`;
 [ $? -eq 0 ]   && systemctl set-environment _WSREP_START_POSITION=$VAR || exit 1 (code=exited, status=0/SUCCESS)
    Process: 1890 ExecStartPost=/bin/sh -c systemctl unset-environment _WSREP_START_POSITION (code=exited, status=0/SUCC
ESS)
    Process: 1892 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0/SUCCESS)
   Main PID: 1878 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 10 (limit: 7308)
     Memory: 112.8M
        CPU: 475ms
     CGroup: /system.slice/mariadb.service
             └─1878 /usr/sbin/mariadbd

Apr 20 18:32:30 ip-192-168-0-163 mariadbd[1878]: 2026-04-20 18:32:30 0 [Note] InnoDB: Loading buffer pool(s) fro…er_pool
Apr 20 18:32:30 ip-192-168-0-163 mariadbd[1878]: 2026-04-20 18:32:30 0 [Note] Plugin 'FEEDBACK' is disabled.
Apr 20 18:32:30 ip-192-168-0-163 mariadbd[1878]: 2026-04-20 18:32:30 0 [Warning] You need to use --log-bin to ma…s work.
Apr 20 18:32:30 ip-192-168-0-163 mariadbd[1878]: 2026-04-20 18:32:30 0 [Note] Server socket created on IP: '0.0.…'3306'.
Apr 20 18:32:30 ip-192-168-0-163 mariadbd[1878]: 2026-04-20 18:32:30 0 [Note] InnoDB: Buffer pool(s) load comple…8:32:30
Apr 20 18:32:30 ip-192-168-0-163 mariadbd[1878]: 2026-04-20 18:32:30 0 [Note] /usr/sbin/mariadbd: ready for connections.
Apr 20 18:32:30 ip-192-168-0-163 mariadbd[1878]: Version: '10.11.14-MariaDB-0+deb12u2'  socket: '/run/mysqld/mys…bian 12
Apr 20 18:32:30 ip-192-168-0-163 systemd[1]: Started mariadb.service - MariaDB 10.11.14 database server.
Apr 20 18:32:30 ip-192-168-0-163 /etc/mysql/debian-start[1894]: Upgrading MySQL tables if necessary.
Apr 20 18:32:30 ip-192-168-0-163 /etc/mysql/debian-start[1913]: Triggering myisam-recover for all MyISAM tables an…ables
Hint: Some lines were ellipsized, use -l to show in full.
admin@ip-192-168-0-163:~$
```

Ključni preizkus zasebne VPC povezljivosti sem izvedel iz EC2-1: ukaz `mariadb -h 192.168.0.163 -u urban -purban AlmaMater -e "SELECT * FROM nakup;"` je uspešno vrnil obe testni vrstici. To potrjuje, da spletni strežnik dosega podatkovno bazo izključno prek zasebnega naslova znotraj VPC omrežja, varnostna skupina vaja07-sg-db pa dovoli TCP 3306 samo iz vira vaja07-sg-web.

```
admin@ip-192-168-0-8:~$ mariadb -h 192.168.0.163 -u urban -purban AlmaMater -e "SELECT * FROM nakup;"
+----+---------+----------+
| id | element | kolicina |
+----+---------+----------+
|  1 | kruh    |        2 |
|  2 | mleko   |        1 |
+----+---------+----------+
```

5. Demo podatkovni strežnik (EC2-3)

Pri EC2-3 sem želel pokazati, kako lahko strežnik namestimo že pri izdelavi instance z `--user-data` parametrom. V skripti `user-data-ec2-db-demo.sh` sem napisal osnovne ukaze `apt-get update`, `apt-get install -y mariadb-server` in `systemctl enable --now mariadb`. Cloud-init je skripto samodejno izvedel ob prvem zagonu instance, kar sem po nekaj minutah potrdil z `cloud-init status --wait` (status: done).

Preko EC2-1 kot bastion sem se povezal na EC2-3 (192.168.0.198) in z `apt list --installed` preveril, da je paket `mariadb-server` dejansko prisoten na sistemu.

```
admin@ip-192-168-0-8:~$ ssh -i ~/vaja07-key.pem admin@192.168.0.198
The authenticity of host '192.168.0.198 (192.168.0.198)' can't be established.
ED25519 key fingerprint is SHA256:XAtP5CIiEZKskfhFc2GgyYKQCjGeNcMud1ZG29qEFuE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.198' (ED25519) to the list of known hosts.
Linux ip-192-168-0-198 6.1.0-44-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.164-1 (2026-03-09) x86_64

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permitted by applicable law.
admin@ip-192-168-0-198:~$ apt list --installed 2>/dev/null | grep mariadb-server
mariadb-server-core/oldstable,now 1:10.11.14-0+deb12u2 amd64 [installed,automatic]
mariadb-server/oldstable,now 1:10.11.14-0+deb12u2 amd64 [installed]
admin@ip-192-168-0-198:~$
```

Z `systemctl status mariadb --no-pager` sem nato potrdil, da servis teče in je aktiven — enako stanje kot pri EC2-2, vendar brez mojih posegov.

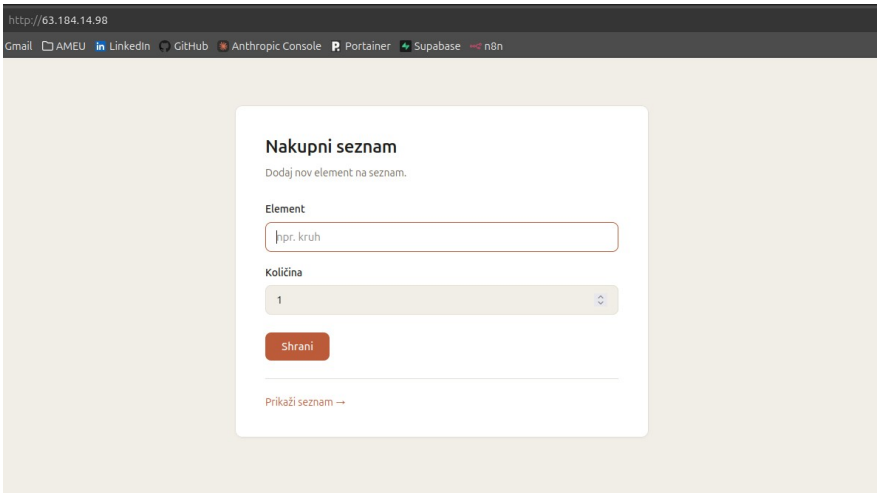
```
admin@ip-192-168-0-198:~$ apt list --installed 2>/dev/null | grep mariadb-server
mariadb-server-core/oldstable,now 1:10.11.14-0+deb12u2 amd64 [installed,automatic]
mariadb-server/oldstable,now 1:10.11.14-0+deb12u2 amd64 [installed]
admin@ip-192-168-0-198:~$ systemctl status mariadb --no-pager
● mariadb.service - MariaDB 10.11.14 database server
     Loaded: loaded (/lib/systemd/system/mariadb.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-04-20 18:00:49 UTC; 40min ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
   Main PID: 1535 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 8 (limit: 7308)
     Memory: 111.7M
        CPU: 830ms
     CGroup: /system.slice/mariadb.service
             └─1535 /usr/sbin/mariadbd

Apr 20 18:00:49 ip-192-168-0-198 mariadbd[1535]: 2026-04-20 18:00:49 0 [Note] Plugin 'FEEDBACK' is disabled.
Apr 20 18:00:49 ip-192-168-0-198 mariadbd[1535]: 2026-04-20 18:00:49 0 [Warning] You need to use --log-bin to ma…s work.
Apr 20 18:00:49 ip-192-168-0-198 mariadbd[1535]: 2026-04-20 18:00:49 0 [Note] Server socket created on IP: '127.…'3306'.
Apr 20 18:00:49 ip-192-168-0-198 mariadbd[1535]: 2026-04-20 18:00:49 0 [Note] InnoDB: Buffer pool(s) load comple…8:00:49
Apr 20 18:00:49 ip-192-168-0-198 mariadbd[1535]: 2026-04-20 18:00:49 0 [Note] /usr/sbin/mariadbd: ready for connections.
Apr 20 18:00:49 ip-192-168-0-198 mariadbd[1535]: Version: '10.11.14-MariaDB-0+deb12u2'  socket: '/run/mysqld/mys…bian 12
Apr 20 18:00:49 ip-192-168-0-198 systemd[1]: Started mariadb.service - MariaDB 10.11.14 database server.
Apr 20 18:00:49 ip-192-168-0-198 /etc/mysql/debian-start[1550]: Upgrading MySQL tables if necessary.
Apr 20 18:00:49 ip-192-168-0-198 /etc/mysql/debian-start[1561]: Checking for insecure root accounts.
Apr 20 18:00:49 ip-192-168-0-198 /etc/mysql/debian-start[1565]: Triggering myisam-recover for all MyISAM tables an…ables
Hint: Some lines were ellipsized, use -l to show in full.
admin@ip-192-168-0-198:~$
```

6. Spletna aplikacija — nakupni seznam

Spletna aplikacija je sestavljena iz štirih datotek v mapi `/var/www/html/`. Datoteka `index.html` prikazuje obrazec z dvema poljema — `element` in `kolicina` — in gumbom Shrani. Datoteka `vstavi.php` prevzame POST podatke, jih preveri in z uporabo prepared statementa vstavi novo vrstico v tabelo nakup. Datoteka `izpis.php` prebere vse vrstice iz tabele in jih prikaže v HTML tabeli. Vse tri strani vključujejo skupno `db.php`, kjer je shranjena konfiguracija povezave, in skupen `styles.css` za enoten videz.

Po odprtju naslova http://63.184.14.98/ v brskalniku se je prikazal obrazec v čisti obliki z belo kartico na kremnem ozadju.



Vnesel sem testni element čokolada s količino 5 in kliknil Shrani. Stran `vstavi.php` je potrdila uspešen zapis v bazo s poudarkom imena elementa in količine.

Po kliku na povezavo Prikaži seznam sem odprl `izpis.php`, ki je prikazal tabelo z vsemi tremi vrsticami: dvema testnima (kruh, mleko) in novo dodano (čokolada). S tem je dokazano, da se podatki pravilno zapišejo v bazo na EC2-2 prek zasebnega VPC omrežja in so ob naslednjem branju pravilno izpisani.

REZULTAT

Vzpostavljeno infrastrukturo sestavljajo en VPC s tremi podomrežji v treh različnih Availability Zone, ena javno dostopna EC2 instanca s spletnim strežnikom ter dve privatno nameščeni EC2 instanci s podatkovnim strežnikom MariaDB. Spletna aplikacija na http://63.184.14.98/ uporabniku omogoča vnos in pregled elementov nakupnega seznama, pri čemer povezava na bazo AlmaMater poteka izključno prek zasebnega naslova 192.168.0.163 znotraj VPC-ja. Demonstracijska instanca EC2-3 je MariaDB dobila samodejno preko user-data skripte ob prvem zagonu, kar pokaže možnost popolnoma avtomatizirane postavitve novih virov.

ČIŠČENJE

Po končanem delu sem zagnal `bash stop-vaja07-ec2.sh`, ki je vse tri EC2 instance spravil v stanje stopped. VPC vaja07-vpc, obe podomrežji, obe route tables, obe varnostni skupini, ključni par in Elastic IP ostajajo, saj bomo delo nadaljevali pri VAJA-08. Po oddaji naloge sem v AWS IAM konzoli deaktiviral in izbrisal access key ter počistil datoteko `~/.aws/credentials` na lokalnem računalniku.